Results for Eedi

theta_m=0.8, theta_f=0.8, rec=50 | N=5,10,...,50 | Ep thresholds 0.4, 0.5, 0.6, 0.7

Each table ranks models within the same graph and student split, using unrounded values.
Bold = best; underline = second best. Ties at the best value are all bold.

Experimental groups

Group	Models	exfr train edges	Test students
1. Original student split; remove zero-forgetting edges	8/8	532,598	234
2. Revised student split (20261007); complete forgetting graph	8/8	664,548	234
3. Revised student split (20261007); remove zero-forgetting edges	8/8	527,549	234

Model columns

2CKG4ER, No-mastery, No-forgetting, No-student-feature, No-exercise-feature, ID-based only

TransE and TransE-adv appear only in groups with eight completed models.

Source locations

Uploaded Eedi ZIP archive (20261007_233001)

Data_Fin_New/_resplit_seed20261007/_full_graph_runs/ + _full_graph_comparison_runs/

Data_Fin_New/_resplit_seed20261007/_forgetting_edge_runs/ + _comparison_runs/

Group 1: Original student split; remove zero-forgetting edges

Train edges: mastery 39,957; forgetting 532,598; recommendation 35,050 | 8/8 models complete

(a) Adaptability (Ada)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only	TransE	TransE-adv
5	0.662	0.657	**0.693**	0.664	0.663	0.662	0.653	0.660
10	0.664	0.656	**0.685**	0.663	0.663	0.662	0.653	0.660
15	0.664	0.655	**0.680**	0.662	0.663	0.661	0.653	0.660
20	0.664	0.654	**0.676**	0.662	0.664	0.660	0.654	0.660
25	0.664	0.654	**0.673**	0.662	0.663	0.659	0.654	0.659
30	0.664	0.654	**0.672**	0.662	0.663	0.659	0.655	0.659
35	0.664	0.654	**0.670**	0.661	0.663	0.659	0.656	0.659
40	0.665	0.655	**0.669**	0.661	0.663	0.659	0.656	0.659
45	0.664	0.655	**0.669**	0.661	0.662	0.659	0.656	0.659
50	0.664	0.655	**0.668**	0.661	0.662	0.659	0.656	0.659

(b) Novelty (NOV)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only	TransE	TransE-adv
5	0.984	0.985	0.987	0.980	0.984	0.982	0.996	**0.997**
10	0.984	0.985	0.983	0.982	0.982	0.979	0.996	**0.997**
15	0.981	0.985	0.980	0.981	0.983	0.979	0.997	**0.998**
20	0.982	0.985	0.978	0.981	0.983	0.979	0.997	**0.998**
25	0.982	0.984	0.977	0.982	0.983	0.979	0.997	**0.997**
30	0.982	0.984	0.976	0.981	0.982	0.979	0.997	**0.997**
35	0.982	0.984	0.977	0.981	0.981	0.979	0.996	**0.997**
40	0.982	0.984	0.978	0.980	0.981	0.979	0.996	**0.997**
45	0.981	0.984	0.979	0.980	0.982	0.978	0.995	**0.997**
50	0.982	0.984	0.980	0.979	0.982	0.979	0.995	**0.997**

Seed 2024; one run per model. Higher is better; no cross-seed standard deviation.

Group 1: Original student split; remove zero-forgetting edges

Train edges: mastery 39,957; forgetting 532,598; recommendation 35,050 | 8/8 models complete

(c) Ep, threshold 0.4

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only	TransE	TransE-adv
5	-0.123	**0.226**	0.162	0.061	-0.127	0.073	0.103	0.089
10	-0.026	**0.337**	0.280	0.113	-0.102	0.136	0.142	0.126
15	0.034	0.358	**0.500**	0.146	-0.050	0.174	0.175	0.151
20	0.107	0.361	**0.518**	0.174	0.002	0.194	0.169	0.156
25	0.123	0.362	**0.557**	0.196	0.014	0.220	0.172	0.175
30	0.133	0.295	**0.600**	0.204	0.000	0.242	0.173	0.191
35	0.122	0.267	**0.573**	0.219	-0.011	0.248	0.169	0.198
40	0.210	0.245	**0.620**	0.229	-0.069	0.277	0.175	0.204
45	0.218	0.214	**0.608**	0.232	-0.036	0.281	0.167	0.221
50	0.301	0.188	**0.602**	0.242	0.034	0.280	0.161	0.225

(d) Ep, threshold 0.5

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only	TransE	TransE-adv
5	-0.197	**0.146**	0.109	-0.003	-0.200	-0.001	0.037	0.017
10	-0.183	**0.209**	0.165	-0.016	-0.250	-0.002	0.034	0.002
15	-0.179	0.192	**0.342**	-0.030	-0.289	-0.022	0.029	-0.017
20	-0.164	0.158	**0.325**	-0.054	-0.311	-0.051	-0.007	-0.042
25	-0.190	0.127	**0.344**	-0.072	-0.343	-0.059	-0.023	-0.059
30	-0.218	0.035	**0.315**	-0.091	-0.396	-0.069	-0.042	-0.061
35	-0.277	-0.016	**0.271**	-0.108	-0.422	-0.091	-0.053	-0.070
40	-0.254	-0.056	**0.299**	-0.117	-0.477	-0.084	-0.055	-0.083
45	-0.256	-0.093	**0.252**	-0.123	-0.457	-0.091	-0.070	-0.090
50	-0.215	-0.144	**0.212**	-0.122	-0.418	-0.103	-0.091	-0.102

Seed 2024; one run per model. Higher is better; no cross-seed standard deviation.

Group 1: Original student split; remove zero-forgetting edges

Train edges: mastery 39,957; forgetting 532,598; recommendation 35,050 | 8/8 models complete

(e) Ep, threshold 0.6

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only	TransE	TransE-adv
5	-0.263	**0.075**	0.057	-0.066	-0.279	-0.067	-0.018	-0.029
10	-0.331	**0.062**	0.035	-0.116	-0.357	-0.124	-0.057	-0.103
15	-0.368	-0.013	**0.168**	-0.189	-0.474	-0.189	-0.097	-0.163
20	-0.383	-0.126	**0.140**	-0.244	-0.535	-0.252	-0.156	-0.218
25	-0.464	-0.202	**0.084**	-0.283	-0.595	-0.290	-0.192	-0.260
30	-0.555	-0.328	**0.043**	-0.328	-0.653	-0.321	-0.229	-0.281
35	-0.648	-0.381	**-0.049**	-0.361	-0.667	-0.351	-0.262	-0.302
40	-0.665	-0.444	**-0.068**	-0.394	-0.709	-0.353	-0.273	-0.316
45	-0.675	-0.506	**-0.140**	-0.410	-0.692	-0.377	-0.298	-0.325
50	-0.647	-0.602	**-0.185**	-0.429	-0.662	-0.398	-0.321	-0.339

(f) Ep, threshold 0.7

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only	TransE	TransE-adv
5	-0.298	-0.014	**-0.004**	-0.143	-0.320	-0.142	-0.074	-0.098
10	-0.435	**-0.109**	-0.117	-0.257	-0.422	-0.257	-0.149	-0.202
15	-0.522	-0.273	**-0.036**	-0.360	-0.547	-0.367	-0.212	-0.286
20	-0.544	-0.438	**-0.101**	-0.435	-0.619	-0.458	-0.294	-0.356
25	-0.638	-0.555	**-0.215**	-0.491	-0.679	-0.511	-0.351	-0.401
30	-0.691	-0.677	**-0.276**	-0.533	-0.736	-0.556	-0.396	-0.432
35	-0.740	-0.706	**-0.415**	-0.563	-0.754	-0.588	-0.430	-0.462
40	-0.738	-0.733	-0.477	-0.589	-0.797	-0.596	**-0.440**	-0.490
45	-0.724	-0.738	-0.574	-0.602	-0.777	-0.631	**-0.468**	-0.506
50	-0.680	-0.781	-0.624	-0.608	-0.742	-0.650	**-0.495**	-0.532

Seed 2024; one run per model. Higher is better; no cross-seed standard deviation.

Group 2: Revised student split (20261007); complete forgetting graph

Train edges: mastery 39,957; forgetting 664,548; recommendation 35,050 | 8/8 models complete

(a) Adaptability (Ada)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only	TransE	TransE-adv
5	0.696	0.690	**0.712**	0.690	0.699	0.686	0.694	0.687
10	0.696	0.689	**0.705**	0.690	0.698	0.687	0.694	0.687
15	0.696	0.687	0.696	0.689	**0.697**	0.687	0.694	0.688
20	0.697	0.687	**0.698**	0.689	0.696	0.687	0.694	0.688
25	0.696	0.687	**0.697**	0.688	0.696	0.687	0.694	0.688
30	0.696	0.687	**0.696**	0.688	0.696	0.687	0.694	0.688
35	0.695	0.686	**0.696**	0.688	0.695	0.687	0.694	0.688
40	0.695	0.686	**0.696**	0.687	0.695	0.687	0.694	0.688
45	0.695	0.685	**0.696**	0.687	0.695	0.688	0.693	0.688
50	0.695	0.685	**0.696**	0.687	0.695	0.688	0.693	0.688

(b) Novelty (NOV)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only	TransE	TransE-adv
5	0.982	0.986	0.984	0.985	0.985	0.984	**0.999**	0.999
10	0.982	0.985	0.981	0.986	0.985	0.986	**0.999**	0.999
15	0.984	0.985	0.984	0.986	0.984	0.985	0.999	**0.999**
20	0.982	0.985	0.986	0.985	0.984	0.984	0.998	**0.999**
25	0.982	0.985	0.986	0.985	0.984	0.984	0.998	**0.999**
30	0.982	0.986	0.986	0.985	0.984	0.984	0.997	**0.999**
35	0.981	0.985	0.987	0.985	0.983	0.984	0.997	**0.999**
40	0.981	0.985	0.987	0.984	0.982	0.983	0.997	**0.999**
45	0.981	0.985	0.987	0.984	0.982	0.983	0.997	**0.999**
50	0.982	0.985	0.987	0.984	0.982	0.983	0.997	**0.999**

Seed 2024; one run per model. Higher is better; no cross-seed standard deviation.

Group 2: Revised student split (20261007); complete forgetting graph

Train edges: mastery 39,957; forgetting 664,548; recommendation 35,050 | 8/8 models complete

(c) Ep, threshold 0.4

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only	TransE	TransE-adv
5	0.005	**0.149**	0.067	0.113	0.054	0.118	0.089	0.097
10	0.139	0.189	0.065	0.182	0.112	**0.192**	0.125	0.172
15	0.180	0.286	**0.343**	0.227	0.180	0.235	0.155	0.232
20	0.211	0.284	**0.424**	0.269	0.318	0.275	0.180	0.268
25	0.244	0.317	**0.500**	0.295	0.291	0.300	0.203	0.290
30	0.365	0.371	**0.553**	0.320	0.292	0.318	0.219	0.317
35	0.393	0.390	**0.468**	0.348	0.246	0.329	0.231	0.333
40	0.449	0.445	**0.544**	0.369	0.297	0.344	0.257	0.355
45	0.465	0.434	**0.538**	0.374	0.353	0.345	0.277	0.364
50	0.354	0.354	**0.564**	0.379	0.393	0.361	0.288	0.371

(d) Ep, threshold 0.5

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only	TransE	TransE-adv
5	-0.070	**0.058**	-0.029	0.029	-0.051	0.033	0.004	0.025
10	-0.024	**0.045**	-0.094	0.030	-0.061	0.034	-0.018	0.027
15	-0.047	0.063	**0.077**	0.017	-0.027	0.028	-0.052	0.029
20	-0.090	-0.021	**0.097**	0.011	0.044	0.030	-0.073	0.022
25	-0.077	-0.043	**0.155**	-0.003	-0.027	0.023	-0.087	0.008
30	-0.045	-0.070	**0.235**	-0.015	-0.067	0.011	-0.099	0.002
35	-0.074	-0.083	**0.186**	-0.016	-0.147	-0.004	-0.128	-0.006
40	-0.080	-0.056	**0.224**	-0.009	-0.133	-0.004	-0.128	0.000
45	-0.090	-0.064	**0.251**	-0.027	-0.126	-0.017	-0.122	-0.014
50	-0.192	-0.127	**0.294**	-0.027	-0.120	-0.013	-0.130	-0.024

Seed 2024; one run per model. Higher is better; no cross-seed standard deviation.

Group 2: Revised student split (20261007); complete forgetting graph

Train edges: mastery 39,957; forgetting 664,548; recommendation 35,050 | 8/8 models complete

(e) Ep, threshold 0.6

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only	TransE	TransE-adv
5	-0.150	**-0.030**	-0.143	-0.040	-0.150	-0.039	-0.067	-0.058
10	-0.190	-0.123	-0.301	-0.107	-0.241	**-0.100**	-0.159	-0.127
15	-0.283	-0.179	**-0.155**	-0.162	-0.291	-0.170	-0.237	-0.187
20	-0.413	-0.329	**-0.155**	-0.209	-0.313	-0.210	-0.286	-0.226
25	-0.486	-0.387	**-0.100**	-0.246	-0.457	-0.253	-0.329	-0.266
30	-0.479	-0.502	**-0.062**	-0.281	-0.545	-0.299	-0.359	-0.311
35	-0.526	-0.609	**-0.119**	-0.313	-0.621	-0.340	-0.399	-0.349
40	-0.541	-0.643	**-0.123**	-0.348	-0.628	-0.362	-0.413	-0.380
45	-0.563	-0.690	**-0.082**	-0.398	-0.647	-0.392	-0.431	-0.417
50	-0.643	-0.769	**-0.082**	-0.434	-0.665	-0.409	-0.454	-0.453

(f) Ep, threshold 0.7

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only	TransE	TransE-adv
5	-0.224	-0.118	-0.228	**-0.116**	-0.223	-0.133	-0.139	-0.119
10	-0.328	-0.291	-0.488	-0.252	-0.382	-0.267	-0.295	**-0.223**
15	-0.497	-0.436	-0.424	-0.365	-0.524	-0.375	-0.419	**-0.317**
20	-0.664	-0.654	-0.504	-0.460	-0.606	-0.454	-0.519	**-0.394**
25	-0.762	-0.728	-0.517	-0.543	-0.767	-0.522	-0.605	**-0.449**
30	-0.759	-0.802	-0.509	-0.607	-0.864	-0.572	-0.673	**-0.505**
35	-0.805	-0.850	-0.615	-0.656	-0.902	-0.618	-0.739	**-0.563**
40	-0.828	-0.861	-0.627	-0.692	-0.892	-0.647	-0.774	**-0.607**
45	-0.854	-0.869	-0.656	-0.733	-0.886	-0.681	-0.802	**-0.653**
50	-0.920	-0.921	-0.704	-0.752	-0.894	-0.711	-0.829	**-0.698**

Seed 2024; one run per model. Higher is better; no cross-seed standard deviation.

Group 3: Revised student split (20261007); remove zero-forgetting edges

Train edges: mastery 39,957; forgetting 527,549; recommendation 35,050 | 8/8 models complete

(a) Adaptability (Ada)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only	TransE	TransE-adv
5	0.700	0.685	**0.712**	0.683	0.699	0.689	0.689	0.691
10	0.700	0.686	**0.705**	0.683	0.698	0.690	0.690	0.692
15	**0.698**	0.684	0.696	0.684	0.697	0.690	0.690	0.691
20	**0.698**	0.683	0.697	0.683	0.697	0.690	0.690	0.691
25	**0.697**	0.682	0.697	0.683	0.697	0.690	0.690	0.691
30	**0.697**	0.682	0.697	0.684	0.696	0.690	0.689	0.692
35	**0.697**	0.682	0.696	0.684	0.696	0.689	0.689	0.691
40	**0.696**	0.682	0.696	0.684	0.696	0.689	0.689	0.691
45	**0.696**	0.683	0.696	0.684	0.695	0.689	0.689	0.691
50	**0.696**	0.683	0.696	0.684	0.695	0.689	0.689	0.690

(b) Novelty (NOV)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only	TransE	TransE-adv
5	0.992	0.986	0.984	0.985	0.985	0.982	0.999	**0.999**
10	0.989	0.988	0.981	0.985	0.987	0.982	**0.999**	0.999
15	0.988	0.987	0.984	0.985	0.986	0.981	**0.999**	0.999
20	0.988	0.986	0.985	0.985	0.985	0.981	**0.999**	0.999
25	0.986	0.985	0.986	0.984	0.982	0.981	0.998	**0.998**
30	0.986	0.985	0.986	0.984	0.982	0.982	0.997	**0.998**
35	0.985	0.984	0.987	0.983	0.982	0.981	0.996	**0.997**
40	0.984	0.984	0.987	0.983	0.983	0.982	0.996	**0.996**
45	0.984	0.984	0.987	0.982	0.982	0.982	0.996	**0.996**
50	0.984	0.984	0.986	0.982	0.983	0.982	0.995	**0.996**

Seed 2024; one run per model. Higher is better; no cross-seed standard deviation.

Group 3: Revised student split (20261007); remove zero-forgetting edges

Train edges: mastery 39,957; forgetting 527,549; recommendation 35,050 | 8/8 models complete

(c) Ep, threshold 0.4

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only	TransE	TransE-adv
5	0.003	**0.213**	0.061	0.098	0.020	0.112	0.109	0.112
10	0.102	**0.320**	0.055	0.171	0.102	0.193	0.206	0.180
15	0.268	**0.464**	0.343	0.229	0.157	0.242	0.251	0.222
20	0.237	**0.474**	0.390	0.262	0.183	0.270	0.259	0.247
25	0.298	**0.519**	0.477	0.288	0.249	0.292	0.277	0.262
30	0.251	0.459	**0.530**	0.307	0.237	0.321	0.292	0.278
35	0.289	0.488	**0.505**	0.322	0.341	0.336	0.298	0.292
40	0.338	0.427	**0.553**	0.336	0.429	0.350	0.319	0.289
45	0.410	0.336	**0.565**	0.349	0.491	0.372	0.339	0.294
50	0.461	0.274	**0.511**	0.350	0.465	0.388	0.354	0.323

(d) Ep, threshold 0.5

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only	TransE	TransE-adv
5	-0.068	**0.136**	-0.035	0.020	-0.058	0.028	0.040	0.018
10	-0.047	**0.205**	-0.100	0.027	-0.028	0.050	0.061	0.012
15	0.014	**0.274**	0.086	0.029	-0.021	0.056	0.050	-0.006
20	-0.051	**0.213**	0.095	0.028	-0.068	0.039	0.023	-0.027
25	-0.016	**0.177**	0.162	0.022	-0.032	0.016	0.010	-0.035
30	-0.086	0.080	**0.241**	0.014	-0.098	0.008	0.000	-0.058
35	-0.100	0.062	**0.214**	0.004	-0.019	-0.008	-0.008	-0.061
40	-0.101	-0.014	**0.261**	-0.002	0.016	-0.012	-0.006	-0.083
45	-0.047	-0.081	**0.291**	-0.011	0.030	-0.014	-0.007	-0.095
50	-0.010	-0.144	**0.277**	-0.020	0.002	-0.010	-0.005	-0.084

Seed 2024; one run per model. Higher is better; no cross-seed standard deviation.

Group 3: Revised student split (20261007); remove zero-forgetting edges

Train edges: mastery 39,957; forgetting 527,549; recommendation 35,050 | 8/8 models complete

(e) Ep, threshold 0.6

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only	TransE	TransE-adv
5	-0.126	**0.011**	-0.152	-0.050	-0.121	-0.048	-0.022	-0.060
10	-0.176	**0.032**	-0.312	-0.106	-0.165	-0.083	-0.057	-0.134
15	-0.163	**0.008**	-0.154	-0.170	-0.215	-0.121	-0.099	-0.178
20	-0.294	**-0.124**	-0.155	-0.217	-0.324	-0.183	-0.156	-0.225
25	-0.382	-0.217	**-0.102**	-0.257	-0.400	-0.238	-0.188	-0.274
30	-0.539	-0.368	**-0.049**	-0.291	-0.528	-0.272	-0.216	-0.332
35	-0.571	-0.401	**-0.118**	-0.323	-0.512	-0.307	-0.250	-0.365
40	-0.625	-0.505	**-0.099**	-0.339	-0.558	-0.335	-0.280	-0.404
45	-0.663	-0.595	**-0.106**	-0.359	-0.590	-0.354	-0.294	-0.420
50	-0.671	-0.719	**-0.159**	-0.381	-0.645	-0.365	-0.307	-0.415

(f) Ep, threshold 0.7

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only	TransE	TransE-adv
5	-0.207	**-0.089**	-0.234	-0.126	-0.213	-0.124	-0.099	-0.122
10	-0.318	**-0.132**	-0.499	-0.263	-0.354	-0.252	-0.195	-0.235
15	-0.379	-0.300	-0.414	-0.376	-0.490	-0.370	**-0.284**	-0.331
20	-0.545	-0.523	-0.517	-0.477	-0.637	-0.474	**-0.381**	-0.426
25	-0.634	-0.652	-0.519	-0.566	-0.719	-0.556	**-0.442**	-0.485
30	-0.823	-0.813	**-0.490**	-0.638	-0.812	-0.616	-0.491	-0.548
35	-0.858	-0.826	-0.602	-0.697	-0.748	-0.663	**-0.519**	-0.584
40	-0.891	-0.934	-0.626	-0.743	-0.754	-0.684	**-0.556**	-0.623
45	-0.867	-0.971	-0.660	-0.774	-0.796	-0.705	**-0.569**	-0.645
50	-0.839	-1.004	-0.733	-0.790	-0.824	-0.723	**-0.586**	-0.643

Seed 2024; one run per model. Higher is better; no cross-seed standard deviation.